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Subject: DASC 5305 Data Visualization

Submission: Phase 1 Report

Introduction

For the homework, I used the (region_2.csv) dataset provided by the professor as a subset of the Global Terrorism Database (GTD), containing over 10,000 terrorist incidents from 1970 to 2020 in the Central America & Caribbean regions. The goals here were to explore trends, patterns, attack methods, target types, and severity of casualties to better understand the terrorism dynamics in this region.

The EDA focuses on the following six variables: incident year (iyear), country (country_txt), primary attack type (attacktype1_txt), primary target type (targettype1_txt), number killed (nkill), and number wounded (nwound). I made use of Python in Google Colab for the data cleaning process and for visualizations, while MS Excel was used to build additional charts on a subset of the data provided by the professor.

EDA Process

Data loading and structure

The raw dataset was loaded in Colab, giving 10,386 rows and 135 columns before cleaning. A preview and .info() were used to understand column types, showing a mix of various numeric and categorical variables.

Data cleaning steps

1. Duplicate removal:

The dataset contained 1,233 duplicate rows, which were removed using `df.drop_duplicates()`, reducing the dataset to 9,153 unique incidents. This ensured that counts, distributions, and time trends were not inflated by repeated records and double-counting.

2. Handling unknown values:

In GTD-style data, the value -9 often encodes values such as those that were “unknown” or “not applicable.” For key numeric and indicator variables (nkill, nkillus, nkillter, nwound, nwoundus, nwoundte, INT_LOG, INT_IDEO, INT_MISC, INT_ANY), all -9 values were replaced with NaN to avoid treating “unknown” as a real numeric value in descriptive statistics and visualizations.

3. Checking for Missing Values:

A missing-value summary showed that many higher-order fields (such as claim3, gname3, attacktype3, and claim2) are entirely missing for this region, indicating that tertiary group and claim information are rarely recorded here. Such fields were not used for the EDA process. For the main analytical features (iyear, country_txt, attacktype1_txt, targtype1_txt, nkill, nwound), coverage is much higher as nkill has 7,095 non-missing values and nwound has 6,945 non-missing values.

4. Selection of Features for EDA:

To satisfy the requirement of exploring at least five variables, the following were selected:

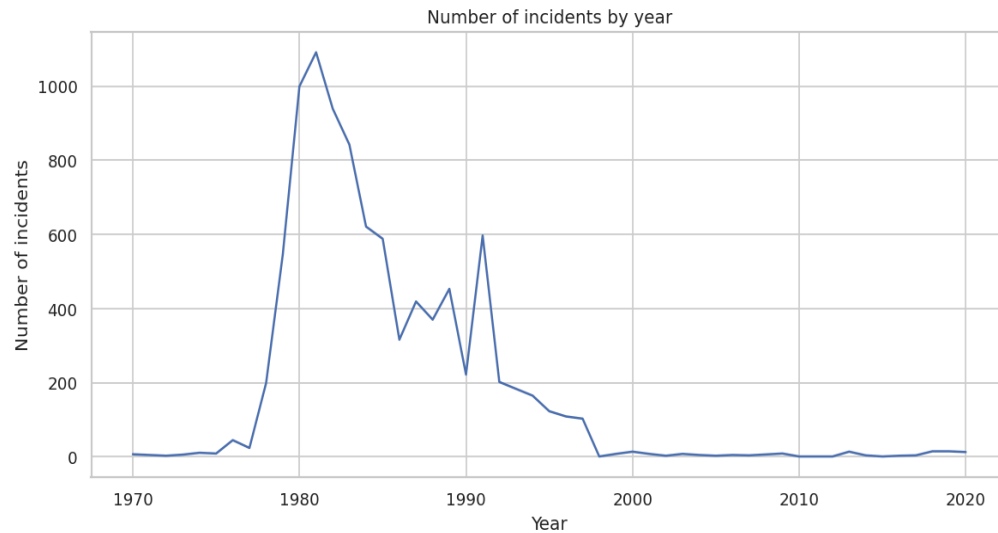
- iyear: year of incident (temporal trends)
- country_txt: country name (geographic distribution)
- attacktype1_txt: primary attack type (tactics that were used)
- targtype1_txt: primary target type (who or what was attacked)
- nkill: number killed (severity of fatalities, if any)
- nwound: number wounded (severity of injuries, if any)

These variables cover when, where, and how severe the dimensions of terrorism in the region are.

Findings and Visual Evidence

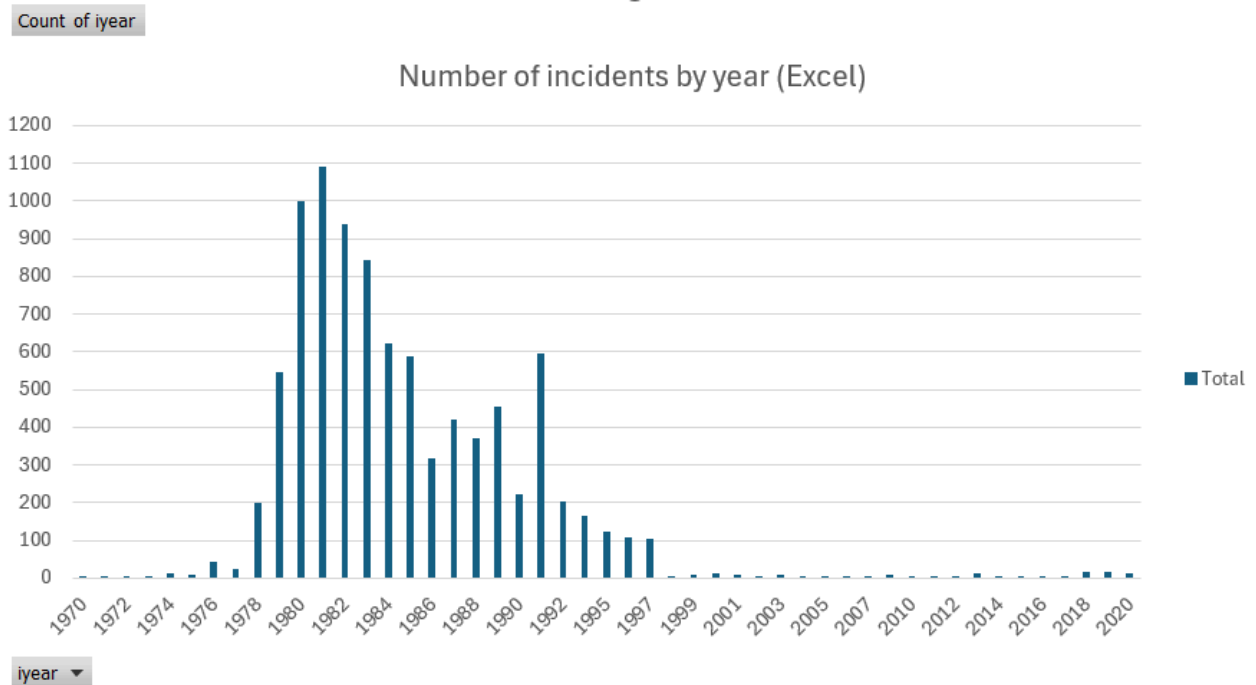
1. Temporal trends: incidents by year

As per the visualizations observed in the colab file, the line chart of incident counts by (iyear) displays a major rise in terrorism during the late 1970s through the mid-1980s, followed by a sharp decline and much lower activity after the early 1990s. The time-series chart validates that incidents start at very low levels in the early 1970s, peak around the early 1980s, and then drop to very low figures in many of the later years.



A Pivot Table built in Excel on the count of incidents by year also gives a column chart with similar patterns, for example: 7 incidents in 1970, 45 in 1976, 199 in 1978, 545 in 1979, 999 in 1980, and 1,091 in 1981. After 1981, the counts declined in a gradual manner, and by 1998, there was only 1 incident, and later years, such as 2010–2012 show 1 incident each, with modest increases again around 2018–2020. This supports the interpretation that terrorism in this region was highly concentrated in a specific historical period rather than evenly spread over time.

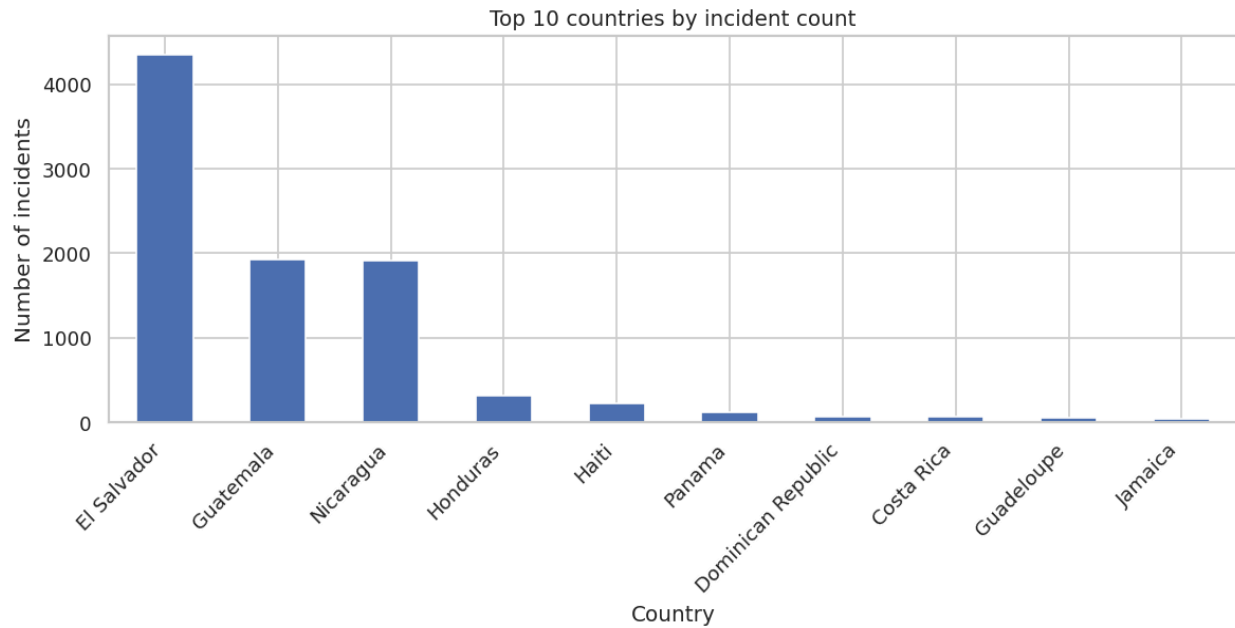
One thing to know is that 2008 does not appear in the year series, both in Python and Excel, because there are no incidents recorded for (iyear = 2008) in this subset.



2. Geographic patterns: countries

Value counts for (country_txt) assert that terrorism in the region of the Caribbean and Central America is dominated by a small number of countries. The top three are El Salvador (4,350 incidents), Guatemala (1,918), and Nicaragua (1,911), followed by Honduras (316), Haiti (224), Panama (118), the Dominican Republic (69), Costa Rica (69), Guadeloupe (50), and Jamaica (34).

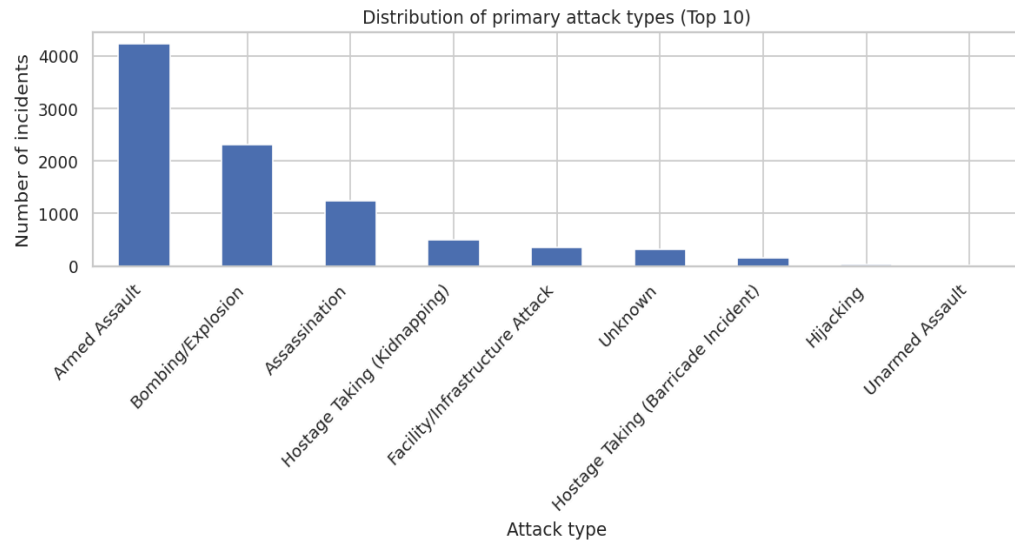
A bar chart of the top 10 countries visually emphasizes how El Salvador alone accounts for nearly half of all terrorist attacks in the provided dataset, with Guatemala and Nicaragua also contributing large shares. In contrast, several other nations in the region have experienced fewer recorded terrorist attacks. This tells us that terrorism during the peak period was heavily concentrated in a very small number of countries experiencing conflicts.



3. Attack types: tactics used

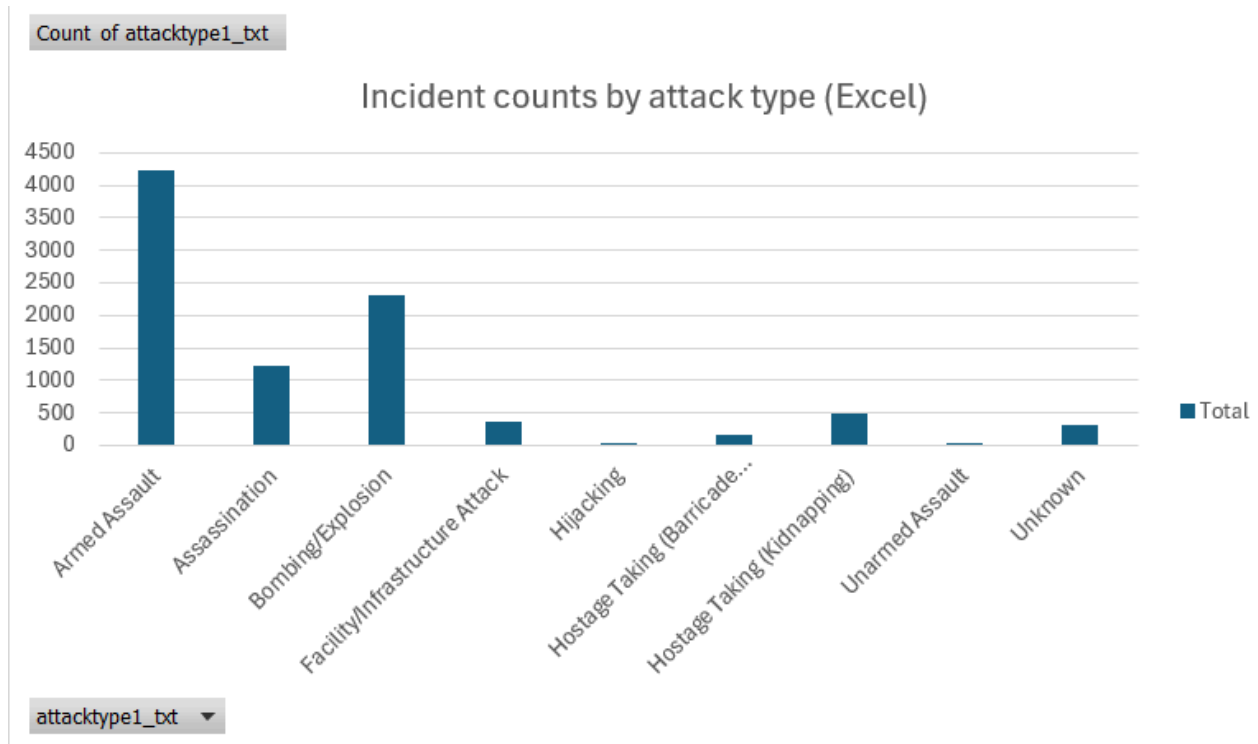
The distribution of (attacktype1_txt) indicates that most incidents in this dataset are either Armed Assault, Bombing/Explosion, or Assassination. Specifically, there are 4,231 Armed Assault incidents, 2,312 Bombing/Explosion incidents, and 1,234 Assassinations; less common types include Hostage Taking (Kidnapping) (503), Facility/Infrastructure Attack (358), Unknown (313), Hostage Taking (Barricade Incident) (160), Hijacking (26), and Unarmed Assault (16).

A bar chart of the top attack types shows Armed Assault clearly dominating, with Bombing/Explosion and Assassination forming a second tier. This implies that in Central America & the Caribbean, day-to-day terrorist activity often involved direct armed confrontations and targeted killings, rather than solely bombings.



In Excel, a PivotTable chart of incident counts by (attacktype1_txt) shows a similar pattern in a more tabular format, reinforcing that a small number of attack types account for the vast majority of such unfortunate events.

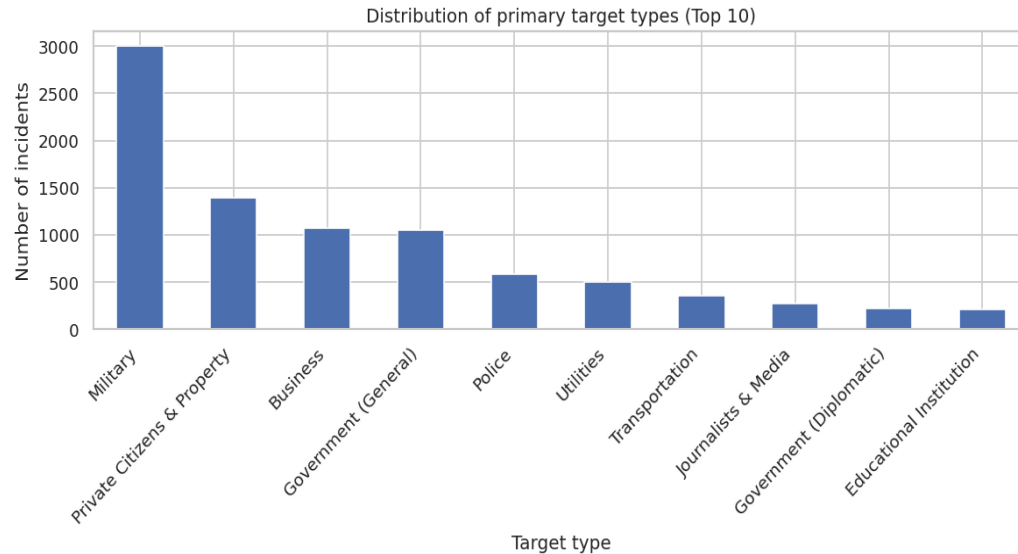
Row Labels	Count of attacktype1_txt
Armed Assault	4231
Assassination	1234
Bombing/Explosion	2312
Facility/Infrastructure Attack	358
Hijacking	26
Hostage Taking (Barricade Incident)	160
Hostage Taking (Kidnapping)	503
Unarmed Assault	16
Unknown	313
Grand Total	9153



4. Target types: who or what is attacked

The distribution of (targettype1_txt) shows that the Military was by far the most attacked, with 3,008 incidents, followed by attacks on Private Citizens & Property (1,387), Businesses (1,071), Government (General) (1,050), and Police (576). Other sectors attacked include Utilities (498), Transportation (348), Journalists & Media (270), Government (Diplomatic) (218), and Educational Institutions (203).

The bar chart for target types makes the military emphasis very clear: attacks on armed forces occur more than twice as often as attacks on any other single category. This suggests that a large share of events fall closer to insurgent or civil conflicts related violence directed at military actors, rather than attacks on civilians alone.

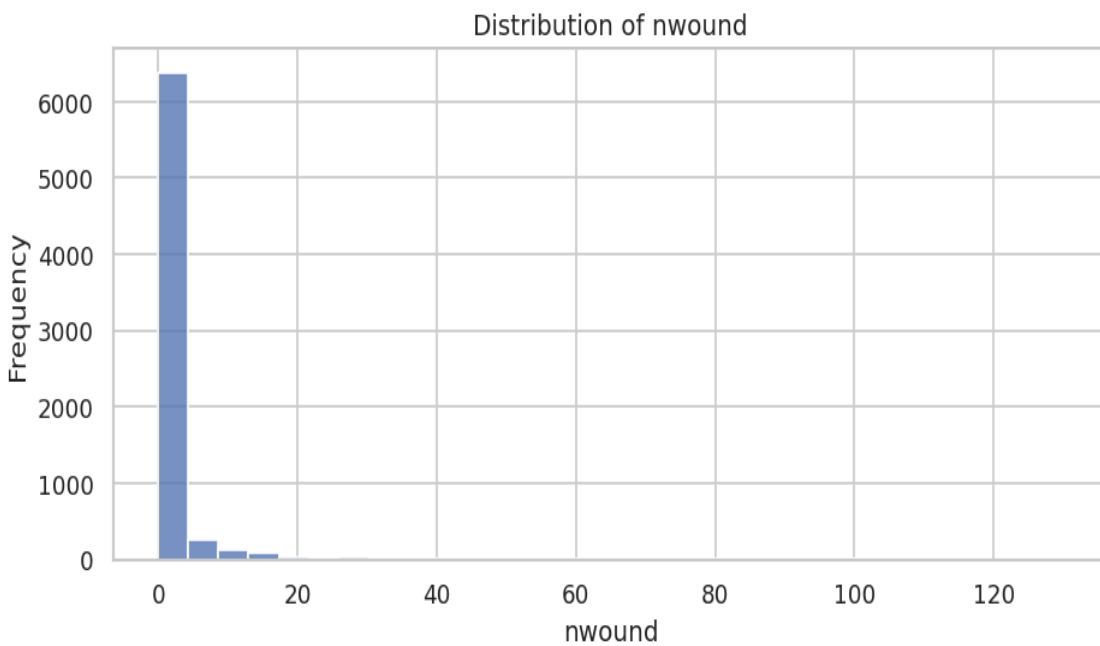
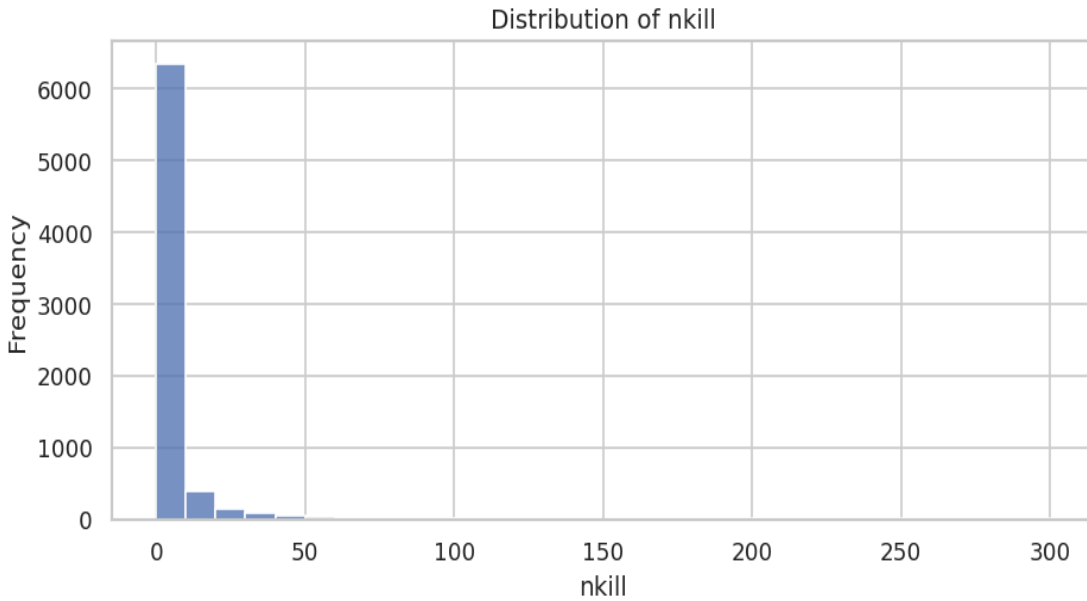


5. Casualties: distributions of nkill and nwound

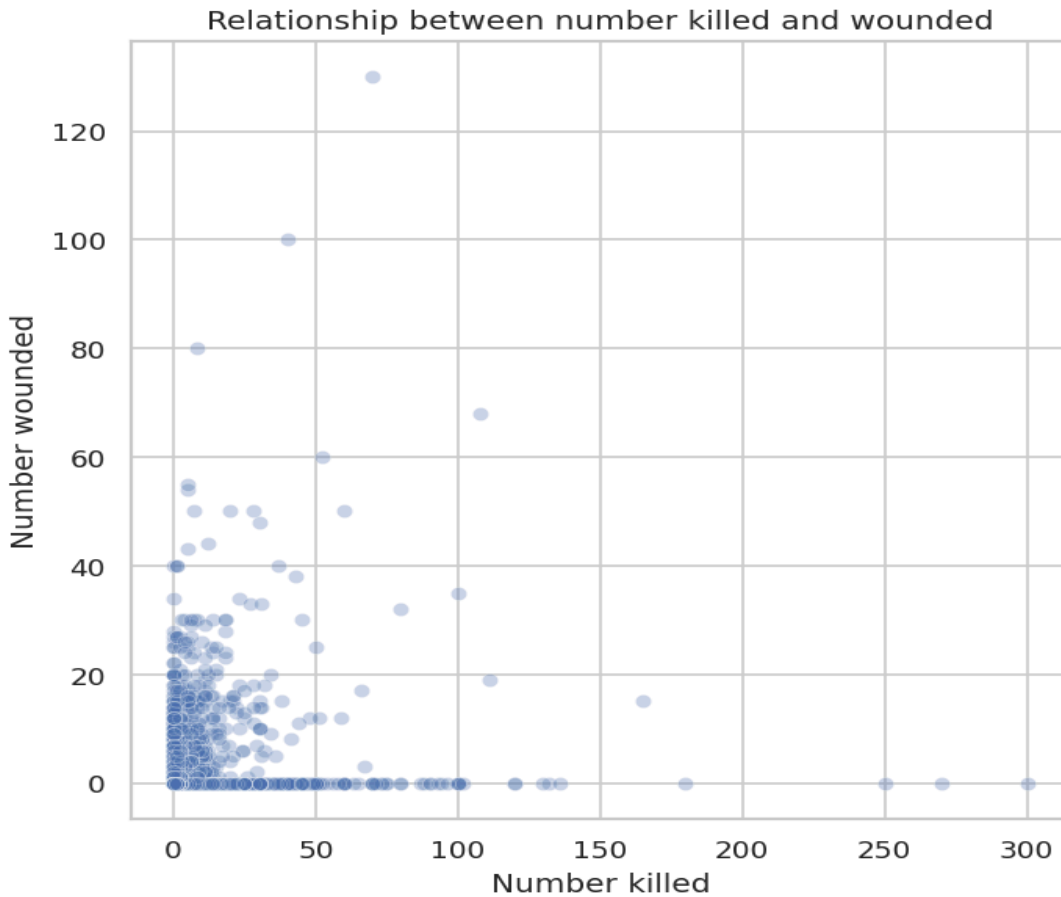
These are the Descriptive statistics for the casualty variables, which show that casualties (fatalities and injuries) are extremely skewed:

- nkill:
 - Count: 7,095 non-missing incidents
 - Mean: nearly 4.03 deaths per incident
 - Median: 1 death
 - 75th percentile: 3 deaths
 - Maximum: 300 deaths in a single incident
- nwound:
 - Count: 6,945 non-missing incidents
 - Mean: 1.29 wounded per incident
 - Median: 0 wounded
 - 75th percentile: 0 wounded
 - Maximum: 130 wounded in a single incident

The histograms for (nkill and nwound) further prove that most of the incidents result in very few casualties or none at all, but a small number of outlier events cause extremely high fatalities and injuries. This is a typical heavy-tailed distribution, and the insights from the EDA process recognize that mean casualties are sparked by a small set of very severe attacks.



The scatterplot of (nkill vs nwound) shows a positive relationship: incidents with many fatalities usually, not always, also have many wounded victims. However, the mass of points is clustered near the origin, which again shows how most events are relatively not that severe in casualty terms.



6. Severity by attack type (Excel averages)

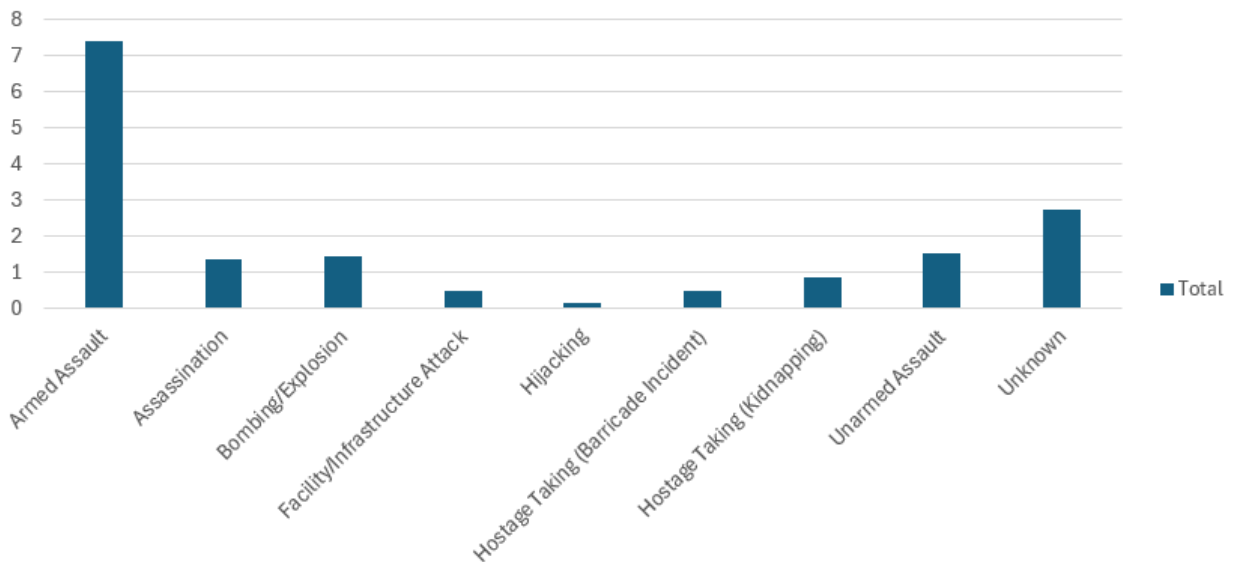
In Excel, a PivotTable was created to calculate the average (nkill by attacktype1_txt), and a column chart visualizes the lethality of each attack type. This supports the frequency chart that while some attack types (like Armed Assault) are very common, others may have higher average fatalities even if they don't occur as frequently.

Using the average fatalities per incident enables us to differentiate between “common but low-severity” tactics and “less common but highly lethal” tactics. Such an approach supports a well-balanced conclusion instead of just simply counting how often each attack type occurs.

Row Labels	Average of nkill
Armed Assault	7.40608
Assassination	1.363488183
Bombing/Explosion	1.459338194
Facility/Infrastructure Attack	0.491666667
Hijacking	0.166666667
Hostage Taking (Barricade Incident)	0.495798319
Hostage Taking (Kidnapping)	0.878205128
Unarmed Assault	1.533333333
Unknown	2.744
Grand Total	4.028611698

Average of nkill

Average nkill by attack type (severity)



attacktype1_bt

Reflection on Tools: Python vs Excel

Both Python and Excel were used to create visualizations for the same core variables, but they offered different strengths.

Python (Colab):

- Python provided a pipeline that could be easily reproduced: the same code cell both cleans the data and regenerates all plots (line charts, bar charts, histograms, scatterplots, and box plots, along with violin plots).
- It handled large data easily and enabled uniform styling across figures, with (seaborn) and (matplotlib) controlling titles, axes, and layouts. It is easier to automate certain processes using Python
- Python was particularly useful for more technical views (such as exploring outliers, combining multiple variables, and creating plots by attack or target type).

Excel:

- Excel, using (region_2_eda_subset.csv), made it very fast to build Pivot Table-based charts such as incidents by year, incidents by attack type, and average fatalities by attack type.
- The drag-and-drop Pivot Table interface made it easy to switch between counts and averages and to quickly prepare the data for presentation.
- Excel charts were easy to export as images and paste directly into a report, but they are less reproducible than Python files and require manual steps to re-create. Basically, there is more manual work in Excel.

Overall, Python offered more flexibility and automation for the exploratory phase, while Excel was great at quickly summarizing and presenting specific views of the cleaned data. Using both tools provided required cross-validation, which is that both the Excel Pivot Tables confirmed the patterns seen in the Python plots, thereby increasing confidence in the results obtained.

Conclusion

This EDA gives a lot of information about terrorism in the Central America and Caribbean regions between 1970 and 2020. Firstly, terrorism is extremely time-concentrated, with a rise in incidents from the late 1970s to the mid 1980s and much lower levels before and after this timeframe. Some years, such as 2008, have no recorded incidents in this subset at all. Secondly, the burden of incidents falls heavily on a few countries, especially El Salvador, Guatemala, and Nicaragua, rather than being uniformly distributed across the region.

Additionally, the most widely utilized tactics are Armed Assault, Bombings or Explosions, and Assassination, and the most common targets are Military forces, followed by Private Citizens & Property, Businesses, and Government entities. This pattern suggests a mix of conflict directed at military institutions, along with violence affecting civilians and economic assets. Moreover, casualty distributions for (nkill and nwound) are highly skewed: most incidents produce zero or low casualties, but a small set of extreme events generates very high fatality and injury counts.

Finally, combining Python and Excel proved effective as Python enabled systematic cleaning and deep exploratory visualizations, while Excel supported quick, easily understandable summary charts thanks to Pivot Tables. Together, they provided a structured, multi-angle understanding of terrorism patterns in this region that can inform further, more focused analyses.